MILITARY SPECIFICATION

FILTER ELEMENT, SHIP FUEL SYSTEM PRE-FILTER, NON-CLEANABLE, HIGH EFFICIENCY

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high efficiency filter elements for use in navy distillate fuel, F-76 conforming to MIL-F-16884 or turbine fuel, aviation grade, JP-5 conforming to MIL-T-5624 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government document.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-T-60 - Tape: Packaging, Waterproof.

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Commend, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL -H-5606 - Hydraulic Fluid Petroleum Base
MIL-T-5624 - Turbine Fuel, Aviation, Grade JP-5
MIL-R-7362 - Rubber, Synthetic, Solid Sheet, Strip and Fabricated Parts, Synthetic Oil Resistant.
MI L- A-8625 - Anodiv Coatings, for Aluminum and Aluminum Alloys
MI L-H-8775 - Hydraulic System Components, Aircraft and Missiles, General Specification for.
MIL-F-8901 - Filter-Separators, Liquid Fuel and Filter-Coalescer Elements, Fluid Pressure: Inspect Ion Requirements and Test Procedures for
MIL-F-15618 - Filter-Separators and Filter Elements, Fluid, Pressure, Aviation and Diesel Fuel, Naval Shipboard.
MIL-F-16884 - Fuel, Naval Distillate
MIL-F-81836 - Filter and Disposable Element, Fluid Pressure, Hydraulic, 3 Micron Absolute.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspect Ion by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Metals, Definition of Dissimilar.
MIL-STD-1188 - Commercial Packaging of Supplier and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications . The following document-s form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B93.18 - Non-Integral Industrial Fluid Power Hydraulic Reservoirs.
B93.19 - Extracting Fluid Samples from the Lines of an Operating Hydraulic Fluid Power System (for Particulate Contamination Analysis), Method for.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

ARP 240 - Determination of Hydraulic Pressure Drop. (DoD adopted)
ARP 785 - Procedure for the Determination of Particle Contamination in Hydraulic Fluids by the Control Filter Gravimetric Procedure. (DoD adopted)

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(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other Informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials. Materials and finishes used in fabrication of the filter element shall not adversely affect or be affected by fuels conforming to MIL-F-16884 or MIL-F-5624, the test fluids included herein, and fuels containing both salt and fresh water.

3.2.1 Metals. Ferrous alloys shall have a chromium content of not less than 12 percent, or shall be protected against corrosion as specified in MIL-H-6775. Copper alloys shall not be used. Aluminum, when used, shall be treated in accordance with MIL-C-5541 or shall be anodized in accordance with MIL-A-8625.

3.2.2 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in direct contact with each other.

3.2.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.4 Fungus proof material. Materials which are not nutrients for fungi shall be used to the greatest extent possible. Where materials are used which are nutrients for fungi, such materials shall be treated with a fungicidal agent.

3.2.5 Imperfections. If resinous or other material is used in the construction of the element for the elimination of imperfections, not more than 5 percent of the filtering area shall be covered by this material.

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3.3 Design and construction.

3.3.1 Seals. Seals shall be synthetic rubber in accordance with MIL-R-7362.

3.3.2 Filter elements. Filter elements shall be as shown on figure 1 and as specified herein. Gaskets shall prevent cocking of the element during installation. Filters shall be the full-flow type rated at 50 gallons per minute (gal/min) with flow through the element from "outside-in". The design shall be such that the filter element cannot be installed in reverse.

3.4 Performance.

3.4.1 Coalescer protection. When tested in series with referee coalescer elements (4.7.3) in a filter/separator and subjected to the tests specified in 4.7.3.1, the pre-filter performance shall be as specified in table I.

TABLE I. Performance of pre-filter and coalescer.

Test time	Maximum filter/separator, lb/in^2 differential	Maximum pre-filter, lb/in^2 differential	Effluent free water content, parts per million
1 hour	2	3	Less than 40
3 hours	5	5	Less than 300

3.4.2 Element pressure drop. The pressure drop through an as received clean element using MIL-H-5606 at rated flow of 50 gal/rein shall not exceed 4 pounds per square inch (lb/in^2) at 100 degrees Fahrenheit (°F) ± 5°F.

3.4.3 Element collapse. The elements shall withstand 180 lb/in^2 differential pressure when tested by the procedure in 4.7.5 without evidence of structural degradation.

3.4.4 Element dirt capacity. Element dirt capacity shall be a minimum of 190 grams of Air Cleaner Fine Test Dust (ACFTD) to a differential pressure of 15 lb/in^2 differential, and 210 grams of ACFTD to a differential pressure of 30 lb/in^2 differential, across the filter element tested by the procedure described in 4.7.6. No clean-up filter shall be used.

3.4.5 Flow fatigue. The elements shall withstand the flow fatigue cycles specified in 4.7.7. Element integrity shall be established by measurement of filtration efficiency ratios (see 4.7.7.2).

3.4.6 Fabrication integrity. The minimum bubble point values shall be established by the filter manufacturer which will yield elements that will fully meet all the requirements of this specification. Elements submitted to the quality conformance bubble test shall meet the minimum values established herein.

3.5 Temperature range. The filter element shall operate throughout a temperature range of 40 to 160°F.

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3.6 Identification of product. Each filter element shall be permanently marked or stamped in accordance with MlIL-STD-130 with the following information:

(a) Military specification number
(b) NSN number
(c) Manufacturer's part number
(d) Manufacturer’s name or trade mark
(e) Non-cleanable

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to pre-scribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor’s overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Inspections The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1 through 4.3.4.

4.3.1 Test fluids. Fuel for tests shall conform to MIL-F-1688. Oil used for tests shall conform to MIL-H-5606.

4.3.1.1 Test fluid filtration. Test fluid conforming to MIL-H-5606 shall be filtered through a 3 micrometer absolute filter meeting the filtration levels specified in MIL-F-81836. Test fuel shall be circulated through a clean-up filter until the fuel contains no visible free water and not more than 2.0 milligrams per liter (mg/L) of solids.

4.3.1.2 Test fluid temperature. Temperature of the test fluid shall be 100 ± 5°F.

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4.3.2 Temperature. Tests shall be conducted at room temperature (70 to 90°F), measured within 12 inches of the test sample.

4.3.3 Instrument accuracy. Flowmeters and gauges shall be accurate within 1.5 percent. They shall be calibrated to determine accuracy.

4.3.4 Immersion. Filter elements shall be immersed in fuel for a period of 72 hours at a fluid temperature of 135°F. All internal parts shall be in contact with the test fluid during this period. The elements shall remain in this fluid at room temperature until further tests are conducted. Prior to testing in oil, the elements shall be thoroughly air-dried for a minimum of 24 hours.

4.4 First article inspection. Four sample elements shall be subjected to the examinations and tests as specified in table II.

TABLE II. First article inspection.

Element No 1 (2nd highest bubble point)	Test paragraph
Examination	4.6
Fabrication integrity (bubble point)	4.7.1
Flow fatigue	4.7.7
Filtration ratio	4.7.4
Element No 2 (highest bubble point)	
Examination	4.6
Fabrication integrity (bubble point)	4.7.1
Pressure buildup (dirt capacity)	4.7.6
Bubble point	4.7.1
Element No. 3 (lowest bubble point)	
Examination	4.6
Fabrication integrity (bubble point)	4.7.1
Filtration ratio	4.7.4
Collapse test	4.7.5
Modified filtration ratio	4.7.5.1(e)
Element No. 4 (2nd lowest bubble point)	
Examination	4.6
Fabrication integrity (bubble point)	4.7.1
Coalescer protection	4.7.3

4.4.1 First article inspection report. The contractor shall prepare a first article inspection report in accordance with the data ordering document included in the contract or order (see 6.2.2).

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4.5 Quality conformance inspection. Each filter element shall be examined in accordance with 4.6 and tested in accordance with 4.7.2. For the pressure collapse and build-up test specified in 4.7.5 and 4.7.6, sampling filter elements shall be in accordance with MIL-STD-105, special inspection level S-1.

4.5.1 Noncompliance. Any filter element which fails the examination of 4.6 and test of 4.7.2 shall be rejected. When the filter element fails to pass the tests of 4.7.5 and 4.7.6 the entire lot shall be rejected. All failures of the tested elements shall be reported immediately by telephone or message. Full particulars concerning previous similar failures, the current failures and action taken to correct the defects shall be submitted to the procuring agency in writing (see 6.2.2). For purposes of quality conformance inspection, a lot shall be defined as all filter elements manufactured from one convoluter set-up.

4.6 Examination. Each element shall be examined to determine conformance to the requirements of this specification for design, workmanship, markings, conformance to manufacturer's drawings, and for any visible defects.

4.7 Tests.

4.7.1 Bubble point (first article inspection). Pre-filter elements shall be tested to determine the initial bubble point. The filter element, containing no fluid, shall be installed in a setup similar to figure 2. The fluid level shall be maintained at approximately 1/2 -inch above the top of the filter element. The air pressure, as indicated in inches of water on the manometer, shall be slowly raised by small increments. The element shall be rotated 360 degrees about its longitudinal axis of each increment of air pressure so t-hat the entire filter area can be observed for the appearance of the first bubble The area of greatest porosity is determined by observing the first bubble on the surface of the element; and the manometer reading in inches of water at which this bubble emits from the element shall be recorded. This test shall be accomplished within a period of 10 minutes after immersion in the fixture. The fluid used shall be Solox 190, or equivalent, at 70 ± 5°F filtered through a 0.45 micrometer membrane Millipore filter, or equivalent.

4.7.1.1 Test point values. The minimum allowable initial bubble point value shall be the lowest initial bubble point of the four first article inspection elements and shall be determined prior to further testing. This value shall be indicated on the first article inspection report and filter element drawing (see 6.2.2).

4.7.2 Bubble point (quality conformance). The filter element, containing no fluid, shall be installed in a setup similar to figure 2. The fluid level shall be maintained at approximately l/2-inch above the top of the filter element, The air pressure shall be preset to 0.1 inch of water less than the minimum initial bubble point. The filter element shall be rotated 360 degrees about its longitudinal axis and the entire filter area scanned for the appearance of any bubbles. There shall be no bubbles emerging from the filter, The fluid used shall be Solox 190, or equivalent, at 70 ± 5°F filtered through a 0.45 micrometer membrane Millipore filter, or equivalent.

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4.7.3 Coalescer protection P~e-filter elements shall be installed in the applicable filter case and tested in series with referee coalescer elements as specified in MIL-F-15618 and qualified to MIL-F-8901. Testing shall be in accordance with the following test plan.

4.7.3.1 Water resistance. Water resistance shall be measured as follows:

(a) All pre-filter elements shall be tested for performance in and resistance to water by use of a testing apparatus similar to figure 3. Components shall conform to MIL-F-8901.
(b) F-76 fuel in accordance with MIL-F-16884 shall be circulated through the system at a flow rate equal to the rated flow of the elements to be tested (pre-filter and coalescer).
(c) The test fuel shall be cleaned by filtering to a maximum level of 1.0 mg/L.
(d) Distilled water shall be continuously injected at a rate of 2.0 percent to 3.0 percent by volume of the total fuel flow
(e) ACFTD shall be added to the fuel tank at an average rate of 10 mg/L
(f) The contaminant shall be distributed uniformly by mixing the fuel with an agitator for 30 minutes. The fuel tank shall be continuously agitated during tests.
(g) The temperature of the test fuel shall be maintained at 75 to 85°F.
(h) Total water content shall be determined by Karl Fischer method. The free water is defined as the difference between the total water content of the effluent fuel sample minus the saturation value of the uncontaminated test fuel at the same temperature as the effluent sample.

4.7.3.2 Test procedure . Tests shall be run at the rated flow of the pre-filter element until the pre-filter fails or the coalescer fails or a maximum of 4 hours, whichever comes first. Record pressure differential in lb/in^2 differential across pre- filter and coalescer at 10 minute intervals. Analyze coalescer effluent fuel sample at 10 minute intervals for free water content. Record values of lb/in^2 differential vs. time for pre-filter and coalescer, and effluent free water vs. time. Flow to coalesce shall be at rated flow of the coalescer as specified in MIL-F-15618.

4.7.3.3 Minimum performance. Minimum performance of pre-filter and coalescer element shall be as specified in table I. The pre-filter and coalescer shall not show signs of collapse or breakthrough during the test.

4.7.4 Filtration ratio test. When tested in the order specified in table II, the pre-filter elements shall exhibit time average filtration ratios at 3 micrometers equal to or greater than 100 and 5 micrometers equal to or greater than 500.

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4.7.4.1 Filtration ratio. The filtration ratio (Fμ) is defined as the ratio of the number of particles greater than a given size (μ) in the influent fluid to the number of particles greater than the same size (μ) in the effluent fluid. From the automatic particle counting data, values of filtration ratio (Fμ) are calculated at each counting time interval. The time averaged filtration ratio (Fμ) for each of the given size ranges shall be averaged for the portion of test run at rated flow up to 30 lb/in^2 across the test element, These results shall be plotted on log normal graph paper, with (Fμ) plotted on the log-scale and the micrometer size range (μ) on the normal-scale. The average filtration ratios for any given particle size shall not be lower than the values specified in 4.7.4.

4.7.4.2 Test system. A typical test circuit is shown on figure 4. Reservoirs shall be constructed with a smooth conical bottom that has an included angle of not more than 90 degrees. The oil entering the reservoir shall be diffused below the surfaces of oil in the reservoir. The lines shall be sized such that turbulent flow exists throughout the system. Fluid aeration shall be held to a minimum. An anti-static additive, such as ASA3 shall be used to dissipate static electric charges. Not more than 50 p/m shall be used

4.7.4.3 Element test circuit. Conditions stated herein shall apply to test circuit validation and to element testing.

(a) The clean-up filter assembly filter (1) shall clean the element test circuit to the extent that particles greater than 3 micrometers in size do not exceed 500 particles per 100 milliliters (mL) of test fluid.
(b) The total fluid volume of the element test circuit shall be equal to or less than that volume which would flow in 1 minute at rated flow plus or minus 2 percent. (Example: 48 liters per minute (L/min) multiplied by 1 minute equals 48 plus or minus 1 liter). This volume may be attained by adjusting the reservoir fluid level.
(c) The test stand shall be capable of supplying fluid to the test element at element rated flow.
(d) Turbulent sampling means in accordance with ANSI B93.19 shall be located upstream and downstream of the test element and shall be connected directly to the automatic particle counter Flow to the automatic particle counter shall be high enough to minimize lag time. The portion of the sampling flow not passing through the counter shall be returned to the element test circuit via the by-pass line. Sample flow shall not be interrupted during the test. In no case shall the element test circuit volume vary by more than 15 percent of the required volume during the test.
(e) pressure taps upstream and downstream of the housing shall be in accordance with SAE ARP 24B.

4.7.4.4 Contaminant injection circuit. A typical circuit is shown on figure 4. The cleanup filter assembly shall be capable of cleaning the contaminant inject ion circuit to the extent that particles greater than 10 micrometer in size do not exceed 250 per mL, and are less than 2 percent of the gravimetric values of injection circuit fluid specified in 4.7.4.7. The total fluid volume of the injection system circuit shall be sufficient to contain the fluid volume

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required by 4.7.4.7 and may be adjusted by varying the level of the oil in the reservoir. A turbulent method in accordance with ANSI B93.19 shall be provided for transferring fluid from the contaminant injection circuit to the element test circuit to yield a flow rate of 0.5 ± 0.025 L/min Samples for analysis shall be taken from the point of injection.

4.7.4.5 Sample analysis methods. Automatic particle counters shall be used to determine the number and size distribution of particles in the fluid samples taken from the element circuit. The particle counters shall be calibrated in accordance with ANSI B93.18 for 3 and 5 micrometer sizes.

4.7.4.6 Test contaminant. The contaminant shall be standardized ACFTD and shall be obtained from Naval Air Development Center (NADC), Warminster, PA 18974. The amount of test dust (W) shall be at least 1.2 times the element apparent capacity In milligrams (mg) to assure a sufficient amount of contaminant for the test.

4.7.4.7 Fluid volume of the contaminant injection circuit. The fluid volume of the contaminant injection circuit is related to the amount of contaminant required to load the element and the element rated flow and shall be determined as follows.

$$(\text{liters}) = \frac{(W, \text{mg}) \times (\text{Injection flow, liters/minute})}{(\text{Element rated flow, liters/minute}) \times (\text{Base upstream level mg/liter})}$$

Where:

= Contaminant injection fluid volume, liters
W = Amount of test dust, mg (minimum) (see 4.7.4.6)
Injection flow = 0.5 liters/minute
Base upstream level: 10 mg/llter

Based on these values, the gravimetric level of the contaminant injection circuit is: $\frac{W\ (\text{mg})}{(\text{liters})}$

Note: Injection fluid volume may be increased as desired by increasing the amount of test dust.

4.7.4.8 Test conditions. (a) The test fluid (element test circuit and contaminant injection circuit) shall be oil conforming to MIL-H-5606. (b) The test fluid temperature shall be 100 ± 5°F throughout the test.

4.7.4.9 Element test circuit validation. Adjust the volume of fluid in the element test circuit to the required volume (see 4.7.4.3). Install a filter housing or conduit. Adjust test fluid temperature to 100 ± 5°F. Contaminate the circuit fluid to a calculated gravimetric level of 10 mg/L using ACFTD. Circulate the fluid in the element test circuit at a rate equal to the element flow rate plus or minus 5 percent. Fully open the valve at the upstream and downstream-sample points. Circulate the fluid in the test system for 1 hour. With the automatic-particle counter sensors connected in-line, record cumulative particle counts at 3 and 5 micrometers every 2 minutes, synchronizing the counting periods of the two sensors. The validation test shall be accepted only if:

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(a) There is less than a 5 percent difference in the cumulative particle counts obtained at each corresponding 2 minute inteval in each size range.
(b) The particle counts obtained for a given size from each sample do n&-deviate more than 10 percent from the average particle counts of that size from all samples.
(c) Each recorded cumulative particle counts per mL is within the range specified as follows:

Particle size	3	5
Number of particles	11900-7930	6200-4140

4.7.4.10 Contaminant injection circuit validation. Validate the contaminant injection circuit at the maximum gravimetric level to be used (see 4.7.4.7). Acid the required quantity of ACFTD to the injection circuit fluid. Extract samples after 30, 60, 90, and 120 minutes at the point of injection into the element test circuit. To obtain these samples, fully open valve and collect at least 200 mL in a sample bottle. Return this fluid to the injection reservoir Then without closing the sample valve, collect the sample for analysis. Analyze samples gravimetrically in accordance with SAE ARP 785. The gravimetric level of each sample shall be within 10 percent of the average of the four samples for the validation of the test stand.

4.7.4.11 Test preparation. Adjust the fluid volume of the filter test circuit to the required volume. Calculate and prepare the required amount of test contaminant and adjust the fluid volume of the contaminant injection circuit (see 4.7.4.7). Install the empty filter test housing in the element test circuit. Circulate the fluid in both circuits through the cleanup filter assembly until the required cleanliness levels are attained (see 4.7 4.3 and 4,7.4.4), as indicated by the automatic particle counters. Bypass the cleanup filter assembly and record the pressure drop of the empty test housing at rated flow and stabilize test conditions. Add the required contaminant (see 4.7.4,7) to the injection system and recirculate for 20 minutes to thoroughly disperse the contaminant.

4.7.4.12 Test method. The test method shall be as follows.

(a) The pre-filter element under test shall be installed in the housing (with bypass valve sealed).
(b) The pump shall be started, and rated flow attained by adjusting the variable speed drive. The cleanup filter assembly shall be bypassed during the entire test. Measure and record the clean assembly pressure drop. Calculate and record the clean element pressure drop, which is equal to the clean assembly pressure drop minus the housing pressure drop measured in 4.7,4.11.
(c) Open the sampling valves and start recording particle counts on the automatic particle counter. Particle counters shall be consistent with 4.7.4.5.
(d) Start flow from contaminant injection system to the filter test system. Maintain a flow rate of 0.5 L/min. Measure and record the flow rate. Open valve B and adjust flow so that the total fluid discarded is 0.5 L/min which equals the injection rate.

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The total volume of fluid in the test system shall remain constant. Collect a sample of the contaminant injection fluid at the point of injection.

(e) Start test recording timer. Record particle counts on the automatic counter every 2 minutes. The counting time periods of the two sensors shall be synchronized. Differential pressure across the test filter shall also be recorded at the same time periods.

(f) At 80 percent of specified terminal differential pressure open valve A. Extract a fluid sample of not less than 500 mL on the upstream side of the test filter assembly.

(g) If test is to be terminated at 30 lb/in^2 differential, then when pressure differential across test element reaches 30 lb/in^2 differential, stop timer, stop fluid sampling flow and stop injection flow from the contaminant system. Continue recording differential pressure across the test filter element.

(h) If collapse test is to be run as part of 4.7.5 then continue test to 180 lb/in^2 differential. Particle counts are not necessary for this test phase.

(i) Collect another sample of the contaminant injection fluid at the point of injection upon completion of test.

(j) Analyze gravimetrically per SAE ARP 785 the two samples extracted from the contaminant injection system (see 4.7.4.12 (d) and (i)). Calculate the average of these two gravimetric levels. The gravimetric level for each sample shall be within 10 percent of this average for the test to be valid.

(k) Analyze gravimetrically the fluid sample extracted from the upstream side of the test filter assembly and report in accordance with SAE ARP 785 (see 4.7.4.12 (f)).

4.7.4.13 Filter efficiencies shall be reported for 3 and 5 micrometers particle sizes as filtration ratios (F_μ) defined in 4.7.4.1.

4.7.5 Collapse test.

4.7.5.1 Procedure. The collapse test procedure shall be as follows.

(a) Run filtration ratio test for element no. 3 according to 4.7.4.12 (a) through (f).

(b) At 30 lb/in^2 differential turn off counters, continue to inject contaminant, and increase flow to 70 gal/min.

(c) Maintain increased flow rate until 180 lb/in^2 element differential pressure is reached.

(d) Stop contaminant injection. Reduce flow to 10 gal/min Switch in test circuit clean-up filter.

(e) Continue testing with 10 gal/rein flow, conforming to 4.7.4.12 (c) through (g). Present filtration ratios (see 4.7.4) on particle counts obtained in 4.7.4.12 (c) through (g).

(f) Plot pressure drop versus cumulative weight contaminant added and include in test report.

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4.7.5.2 Criteria for acceptance. The criteria for acceptance shall be as follows:

(a) There shall be no evidence of structural, seal, or filter medium failure.
(b) There shall be no decrease in slope of the pressure drop versus contaminant added curve.
(c) The performance requirements of table II shall be satisfied.

4.7.6 Pressure buildup. The test shall be conducted on element no. 2 at rated flow of the element. A typical test circuit is shown on figure 5.

4.7.6.1 Test procedure. The test procedure shall be as follows:

(a) The test fluid used in the system shall be pre-cleaned as specified in 4.3.1.1. The cleanup filter assembly must be bypassed during testing.
(b) The pre-filter element under test shall be installed in the test housing.
(c) The pump shall be started, valves A and B opened, and rated flow attained by adjusting the varidrive. The differential pressure drop across the element shall not exceed 4.0 lb/in^2 differential.
(d) Standardize ACFTD shall be added through the dust valve H in Increments of 6 grams at 4-minute intervals. Two minutes after each test dust addition, pressure differential flow and temperature shall be recorded. The differential pressure across the element shall not exceed 30 lb/in^2. The pump shall not be stopped during the test.
(e) A curve showing differential pressure across the element versus cumulative weight added in grams of standardized ACFTD shall be included in the test report.

4.7.7 Flow fatigue. The pre-filter element shall be installed in the test housing and shall be subjected to pressure-flow cycles at flow rates specified in 4.7.7.1. A cycle shall consist of increasing the differential pressure across the filter element from zero to the maximum specified and back to zero by first increasing, then decreasing, the flow through the test element, which has been loaded with the minimum amount of ACFTD necessary to achieve the required differential pressures.

4.7.7.1 The cycling rate shall not exceed 300 cycles per minute. The number of cycles and test flow cycles at each differential pressure shall be as follows:

Number of cycles at lb/in^2 differential	Flow ranges (gal/min)
90,000 cycles at 10	15 - 50
8,000 cycles at 15	50 - 100
2,000 cycles at 20	50 - 100
500 cycles at 30	50 - 100
100 cycles at 120	50 - 100

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4.7.7.2 Perform the filtration ratio test as specified in 4.7.4. Reduce flow to 15 gal/min and stabilize downstream count as specified in 4.7.4.3 before restarting the contaminant injection fluid flow. Run time average filtration ratios to 30 lb/in^2 differential in conformance with 4.7.4.12. Filtration efficiency ratios shall be as specified in 4.7.4 and reported for 3 and 5 micrometer particle sizes.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

5.1.1 Level A. Each filter element shall be preserved and packaged by one of the following methods:

(a) Each filter element shall be preserved in accordance with MIL-P-116, method IA-13.
(b) Each filter element shall be inserted in a polyethylene bag fabricated from material conforming to L-P-378, type I, class 1, thickness 0.0040, and closed by heat sealing. The bag containing the element shall be placed in a close- fitting box conforming to PPP-B-636, W6c, style optional. Closure shall be in accordance with method V of the appendix to the box specification The box shall be sealed with tape conforming to PPP-T-60, type IV.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A. The filter elements shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds. Strapping shall conform to QQ-S-781, class I, type I or IV, size as applicable. Unless otherwise specified, (see 6.2.1) strapping shall be finish B. When specified, strapping shall be finish A.

5.2.2 Level B. The filter elements shall be packed together in close-fitting boxes conforming to PPP-B-601, domestic type, style optional; PPP-B-640, class 2, style optional; or PPP-B-636, V3c. The gross weight of each box conforming to PPP-B-601 or PPP-B-640 shall not exceed 200 pounds.

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5.2.3 Commercial. Filter elements, preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The filter elements covered by this specification are intended for use as protection for filter/coalescers qualified to MIL-F-8901 and MIL-F-15618 in ships fuel service system pre-filter assemblies. Fuel service systems operate with either MIL-F-16884 or MIL-T-5624 fuel at operating pressures of 150 lb/in^2 maximum.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) If first article Inspection is required (see 3.1).
(c) Level of preservation, packaging and packing required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27 .410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.1, 4.7.1.1	First article inspection report	DI-T-4902	----
4.5.1	Failure summary report	UDI-R-20612	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000. 19L., Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the

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Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

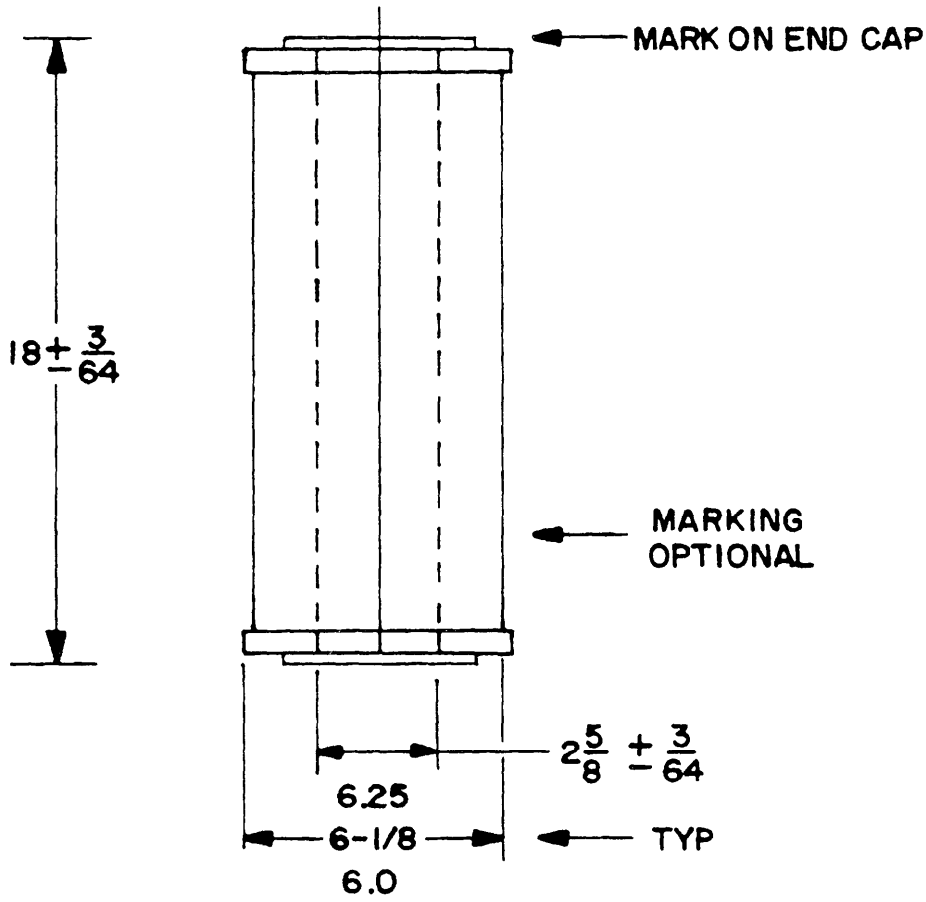
6.3 First article inspection. When a first article inspection is required, the item should be first article sample. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Elements manufactured in excess of contract quantity may be placed in stock and accepted for later contracts (within 3 years) based on prior tests of the specific lot.

Preparing activity:
Navy - SH
(Project 4330-N035)

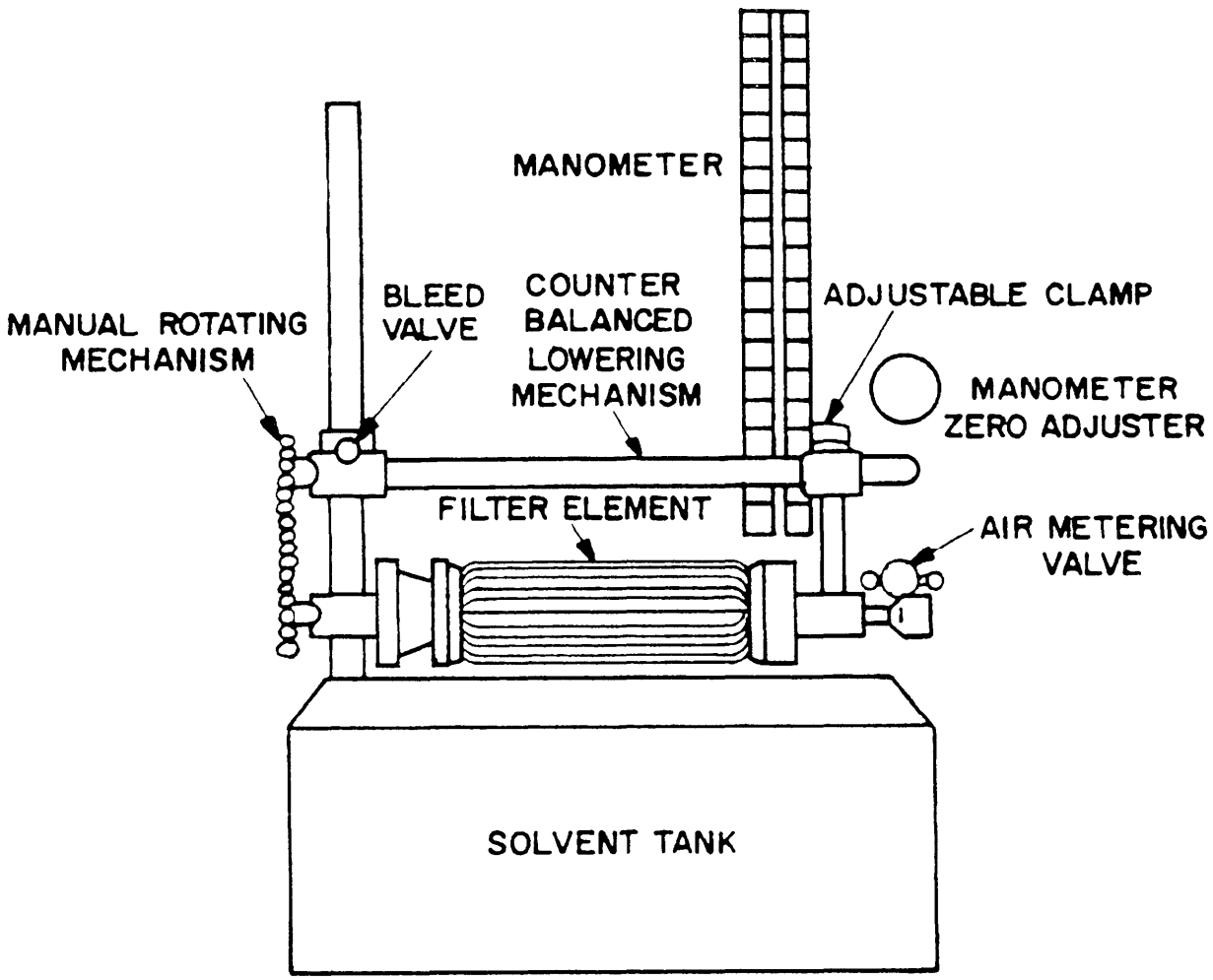
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SH131728

FIGURE 1. Pre-filter element.

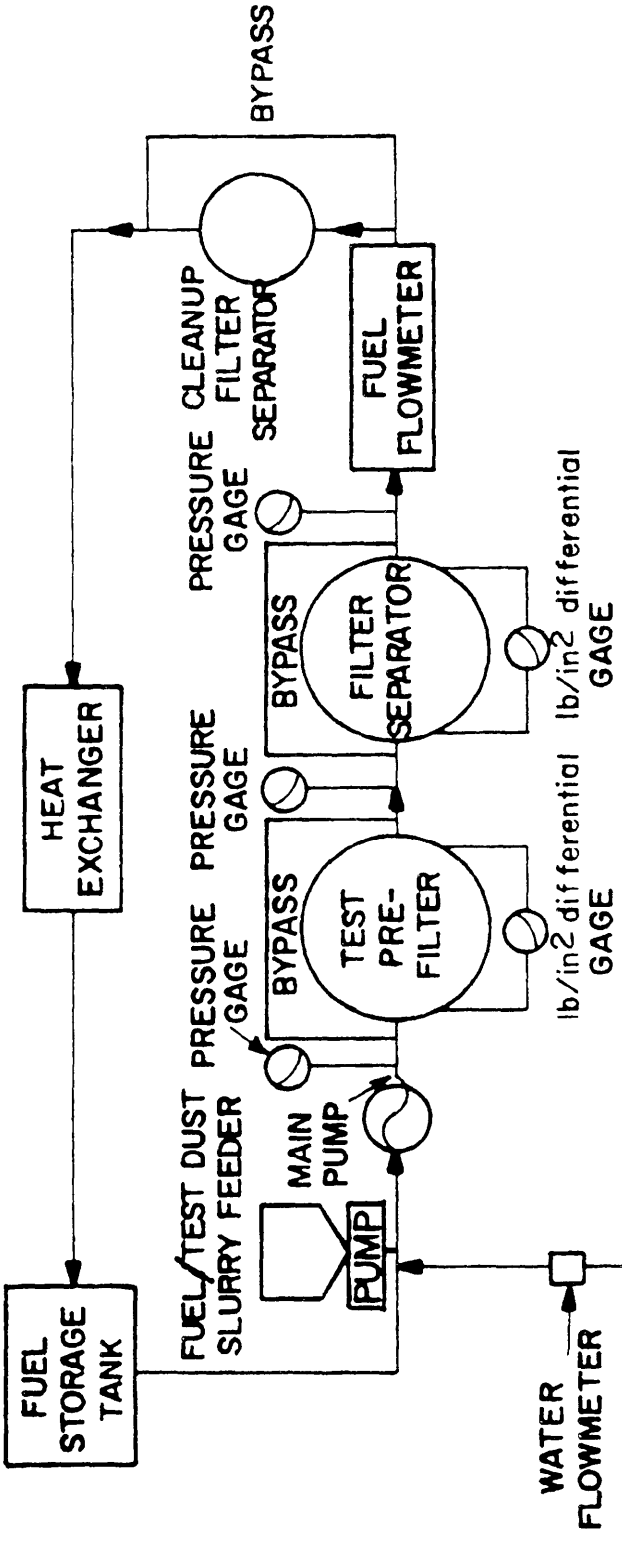
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SH 131729

FIGURE 2. Bubble point

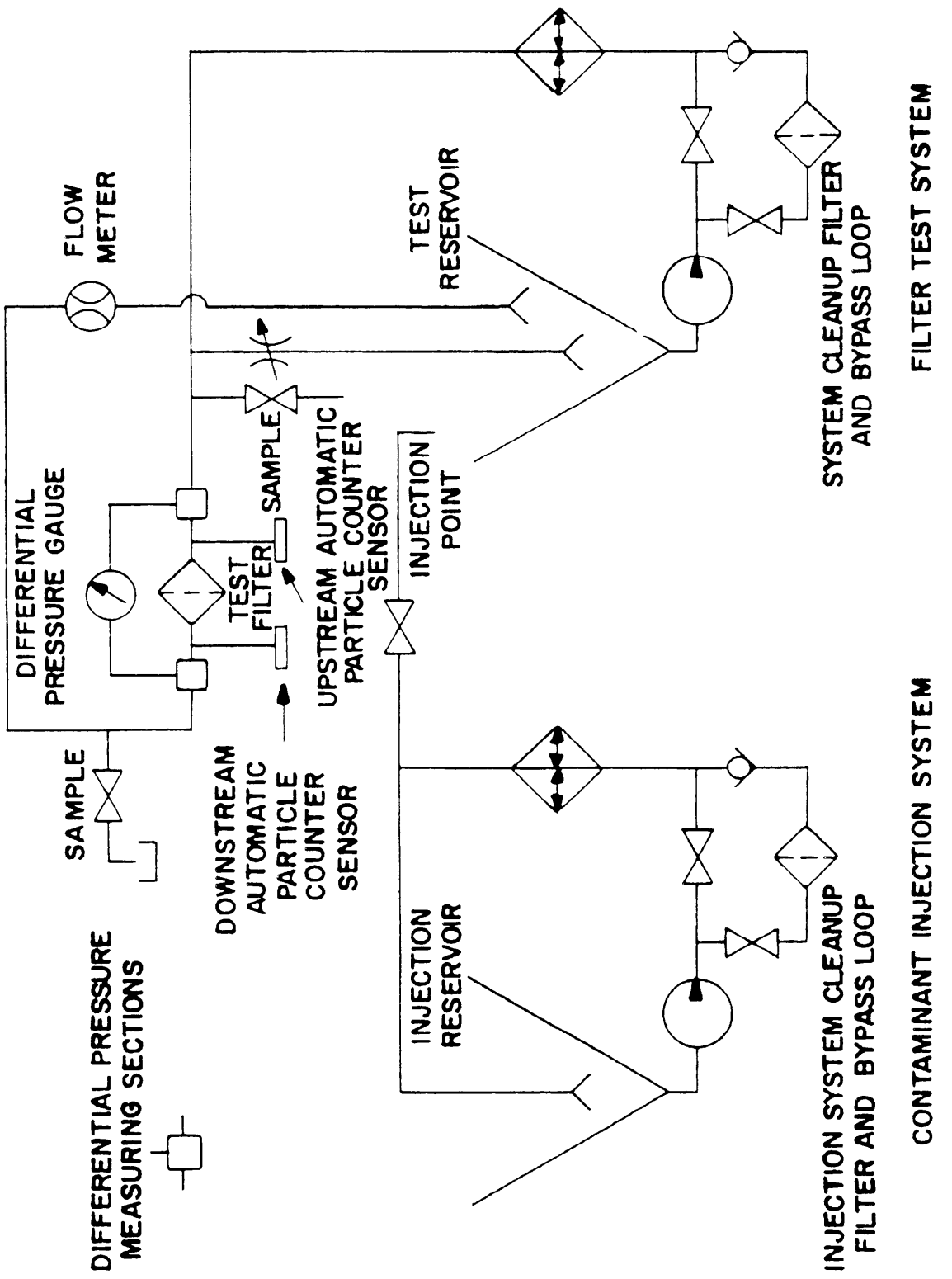
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SH 131730

FIGURE 3. Coalescer protection.

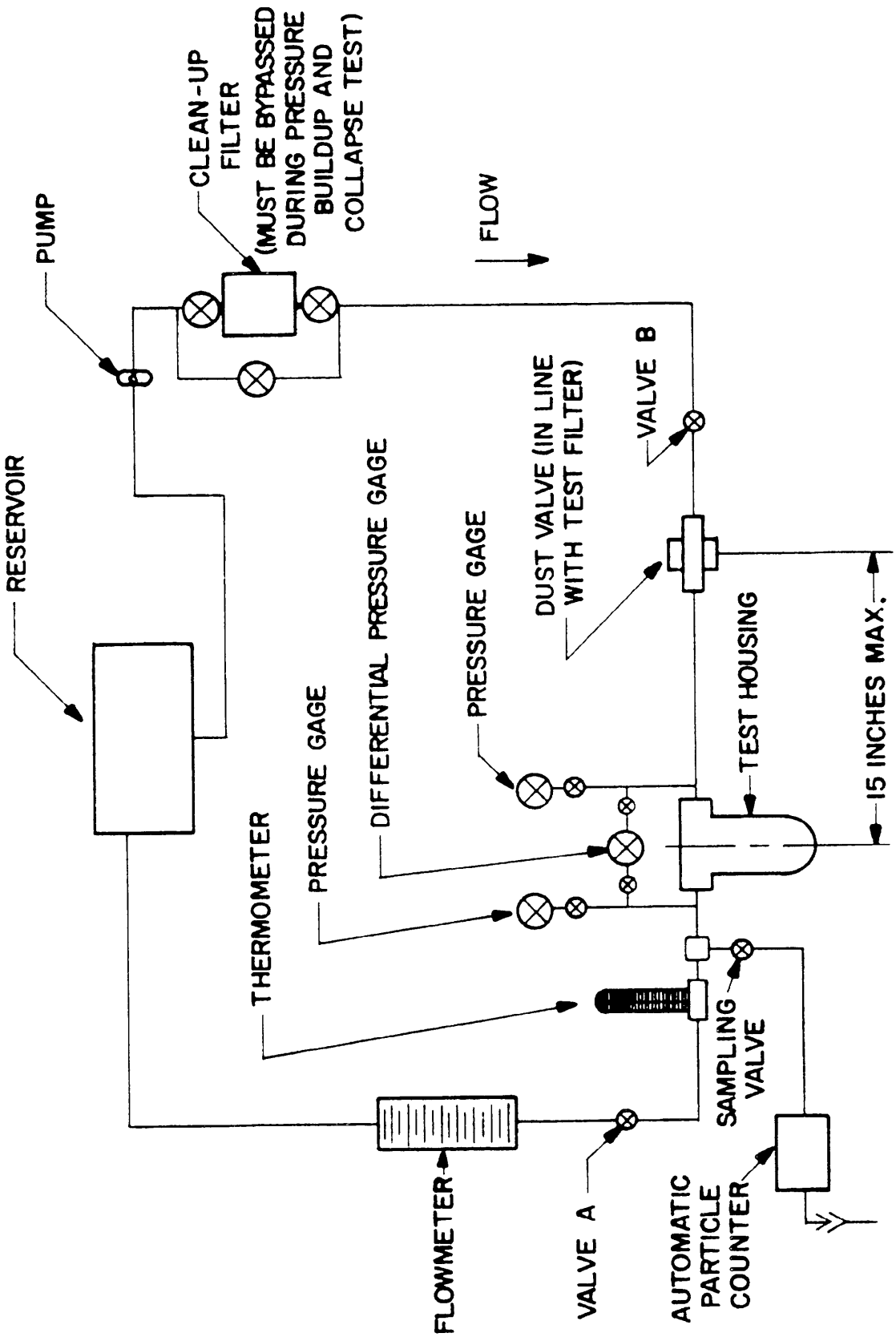
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SH 131731

FIGURE 4. Typical filter performance test circuit.

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FIGURE 5. Dirt capacity.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-24666(SH)	FILTER ELEMENT, SHIP FUEL SYSTEM PRE-FILTER, NON-CLEANABLE, HIGH EFFICIENCY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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